

EDEXCEL IAL CHEMISTRY • UNIT 2 (WCH12)

Unit 2

Energetics, Group Chemistry, Halogenoalkanes and Alcohols

A Complete Revision Guide

Built by analysing every Section A, B and C question and every mark-scheme point from **18 official past papers** (Jan 2020 – Jun 2026, paper code WCH12/01), so that every topic, every recurring question style and every examiner ‘do not award’ trap is covered in one place.

18

past papers analysed

80

marks per paper —
mapped topic by topic

15

chapters covering the
entire specification

100%

of 6-mark extended
questions decoded

Pearson Edexcel International Advanced Subsidiary / Advanced Level — Paper WCH12/01 — 1 hour 30 minutes — 80 marks

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CHAPTER 1

How This Paper Works

Structure, timing, command words and the shape of a full-marks answer

Paper structure at a glance

Every WCH12/01 paper follows an identical, predictable shape. Across all 18 papers analysed for this guide, the paper is always **80 marks in 1 hour 30 minutes** (roughly 1.1 minutes per mark — leave yourself no more than 68 seconds per mark and you finish with time to check calculations), split into three sections:

Section	Format	Typical marks	What it tests
A	17–18 multiple-choice items (some split a/b/c), 1 mark each	20 (fixed every year)	Breadth — a little of everything, including topics too small for Section B/C
B	2–5 structured short-answer questions	38–42	Calculations, definitions, equations, short explanations, mechanisms, graphs
C	2–4 longer structured questions, always including one 6-mark extended-writing (*) question	18–22	Multi-step problems, synthesis routes, and the QER (quality of extended response) question

*Section A is always worth exactly 20 marks; Sections B and C together always total 60 marks. The **6-mark star (*) question always appears once per paper**, in Section B or C, and is always an extended, linked-reasoning answer — see Chapter 14.*

Command words — what the examiner actually wants

Command word	What it requires
State / Give / Identify	A short factual answer — one line, no explanation needed and none rewarded.
Explain	State the fact <i>and</i> give the chemical reason. A bare correct fact with no reasoning typically scores half marks.
Describe	Say what happens/what is observed, in a logical sequence, without necessarily explaining why.
Suggest	A question type you may not have seen before — apply general principles to a novel context. Any chemically sound idea is credited.
Deduce / Determine / Calculate	Show full working. Marks are awarded for individual steps (moles, ratio, final answer) even if the final value is wrong — never leave a calculation blank .
Compare and contrast	State at least one similarity <i>and</i> one difference — marks are lost for only giving differences.
Justify	Your stated answer must be backed by a specific reason, number or comparison, not just repeated.
Discuss / Evaluate	Always the 6-mark QER style — a connected, structured paragraph is required (see Chapter 14).

Marks you must never lose

⚠ COMMON ERROR — State symbols

Any question that says “include state symbols” loses a whole mark for a correct but symbol-less equation. This appears in equation questions almost every single session.

⚠ COMMON ERROR — Significant figures / sign / units

Calculation questions routinely say “give your answer to an appropriate number of significant figures” and/or “include a sign and units”. These are separate, explicitly awarded marking points — a numerically correct answer with the wrong sign, missing units, or given to 5 s.f. from data quoted to 3 s.f. loses marks even though the chemistry is right.

♦ EXAM TIP — “Correct answer, no working” still scores full marks

Nearly every multi-step calculation mark scheme explicitly allows a bare correct final answer to score full marks. But if you make an arithmetic slip, an examiner can only award the method marks if your working is visible — so always show every line.

♦ EXAM TIP — Error carried forward (‘TE’ / transferred error)

If you make an early numerical error but then apply the *correct method* to your own wrong number, most of the later marking points are still awarded (“TE throughout” appears on the majority of multi-step calculations). This means a mistake in step 1 of a 5-mark calculation rarely costs you all 5 marks — keep going, don’t give up on a question because one number looks wrong.

Two golden habits seen on every paper

- 1. Always check units and state symbols before moving on** — they are free, mechanical marks that cost nothing beyond a habit.
- 2. Never leave a numeric answer blank.** Every single paper studied rewards partial working; an empty box scores zero even when the method is simple.

CHAPTER 2

Topic Frequency Map

Every recurring theme across 18 papers, at a glance

The grid below marks, paper by paper, whether each major theme appeared anywhere in that paper (Section A, B or C). It was generated by systematically searching the text of every paper for the relevant keywords and chemistry. Anything showing 16–18/18 is essentially guaranteed to appear on your paper in some form — make sure every box in this guide for that row is second nature before moving on to anything else.

	J20	O20	J21	S21	O21	J22	S22	O22	J23	S23	O23	J24	O24	J25	S25	O25	J26	S26	
Halogenoalkanes	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	18/18
Intermolecular forces / boiling temps	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	18/18
Oxidation numbers	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	18/18
Group 2 thermal stability	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	17/18
Infrared spectra	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	16/18
Equilibrium / Le Chatelier	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	16/18
Calorimetry ($Q = mc\Delta T$)	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	15/18
Chlorine + water / NaOH	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	15/18
Enthalpy definitions ($\Delta_f H$, $\Delta_c H$, $\Delta_a H$...)	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	15/18
Conc. H_2SO_4 + halides	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	14/18
Bond enthalpy data	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	13/18
Hess cycle	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	13/18
Maxwell-Boltzmann	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	13/18
Mechanism drawing (curly arrows)	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	13/18
Silver nitrate tests	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	13/18
Titration	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	13/18
Flame tests	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	12/18
Gas moles (molar volume, $pV = nRT$)	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	12/18
'Appropriate s.f.' demanded	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	12/18
Alcohol oxidation (dichromate)	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	11/18
Disproportionation	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	11/18
Mass spectra	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	11/18
Half-equations / balancing	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	10/18
Rate from graph / tangent	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	8/18
Atom economy	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	6/18
Graph plotting	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	4/18

Presence of each theme across all 18 papers analysed (Jan 2020 – Jun 2026). ■ = present in that paper's Section A, B or C.

How to read this map

- 1. Rows scoring 16-18/18** (intermolecular forces, oxidation numbers, halogenoalkane substitution/elimination, Group 2 thermal stability, equilibrium/Le Chatelier, infrared spectra, chlorine chemistry) are core, non-negotiable topics — they appear on virtually every single paper and are worth building rock-solid first.
- 2. Rows scoring 10-15/18** (calorimetry, enthalpy definitions, bond enthalpies, Hess cycles, redox half-equations, disproportionation, flame tests, conc. H_2SO_4 with halides, silver nitrate tests, titrations, Maxwell-Boltzmann, mechanisms, alcohol oxidation, mass spectra, gas-volume calculations, significant figures) are very likely but not guaranteed on any single sitting — they rotate, so if one hasn't appeared for a few sessions it becomes more likely soon.
- 3. Rows below 10/18** (atom economy as its own question, rate-from-graph tangents, full graph-plotting questions) still appear regularly enough that they cannot be skipped, but are less likely to carry heavy marks in any one sitting.

What this means for your revision order

This guide is ordered so that the highest-frequency, foundational topics come first (energetics, kinetics, equilibrium, redox) because they are used as tools inside almost every other chapter — Group 2 thermal stability questions need oxidation-number and bonding reasoning, halogenoalkane questions need intermolecular-force and mechanism reasoning, and every extended (*) question draws on several chapters at once. Work through the guide in order once, then use Chapter 15's checklist for rapid final review.

CHAPTER 3

Energetics

Enthalpy definitions, Hess's Law, bond enthalpies and calorimetry

3.1 Enthalpy definitions — learn every one word-perfect

Section A almost always includes at least one “which equation represents ...” question, and Section B/C frequently asks you to *state* a definition outright. Every definition below must include: **one mole** of the stated substance, **standard conditions** (100 kPa and a stated temperature, usually 298 K), and **standard states** for all species.

Enthalpy change	Symbol	Definition (learn exactly)
Formation	$\Delta_f H^\ominus$	The enthalpy change when 1 mole of a compound is formed from its elements in their standard states, under standard conditions.
Combustion	$\Delta_c H^\ominus$	The enthalpy change when 1 mole of a substance is completely combusted (burned) in excess oxygen, under standard conditions, all reactants and products in their standard states.
Atomisation	$\Delta_{at} H^\ominus$	The enthalpy change when 1 mole of gaseous atoms is formed from the element in its standard state, under standard conditions.
Neutralisation	$\Delta_{neut} H^\ominus$	The enthalpy change when 1 mole of water is formed from the reaction of an acid with an alkali (or $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$), under standard conditions.
Reaction	$\Delta_r H^\ominus$	The enthalpy change for a reaction as written by the molar quantities shown in the equation — the only one that is <i>not</i> automatically defined as “per mole of substance”, since the amounts depend on the equation's stoichiometry.
Solution	$\Delta_{soln} H$	The enthalpy change when 1 mole of a solute dissolves completely in water (or another solvent), to give an infinitely dilute solution.
⚠ COMMON ERROR Section A repeatedly tests whether you can spot the <i>one wrong equation</i> among four — typical traps: using $1\frac{1}{2} O_2$ or a fractional coefficient in a combustion equation for a species where this gives the wrong product state; showing an element that is <i>not</i> in its standard state (e.g. $Cl(g)$ instead of $\frac{1}{2}Cl_2(g)$ for atomisation, or a gas-phase element where the standard state is solid/liquid); or writing 2 mol of product formed instead of 1 mol. Always check: (1) is it exactly 1 mole of the defined species, (2) are all elements shown in their standard physical state.		

3.2 Hess's Law

Hess's Law states that **the total enthalpy change for a reaction is independent of the route taken**, provided the initial and final conditions are the same. Every Hess-cycle question follows the same three steps:

1. Draw (or complete) the triangle/rectangle with the direct route across the top and the indirect (alternative) route round the sides.
2. Add arrowheads so that every arrow points the way the reaction actually goes — this is itself often a specific mark.
3. Go *around* the cycle: (sum of enthalpy changes on the indirect route) = (enthalpy change on the direct route), being careful with sign when an arrow is being travelled against its own direction.

WORKED EXAMPLE — Formation enthalpy from combustion data

Find $\Delta_f H^\ominus$ for propane, $C_3H_8(g)$, given $\Delta_c H^\ominus[C(s)] = -394 \text{ kJ mol}^{-1}$, $\Delta_c H^\ominus[H_2(g)] = -286 \text{ kJ mol}^{-1}$, $\Delta_c H^\ominus[C_3H_8(g)] = -2220 \text{ kJ mol}^{-1}$.

Route: $3C(s) + 4H_2(g) + 5O_2(g) \xrightarrow{\text{direct, } \Delta_f H} C_3H_8(g) + 5O_2(g)$, and both sides also burn down to $3CO_2(g) + 4H_2O(l)$.

$\Delta_f H = [3 \times (-394) + 4 \times (-286)] - (-2220) = (-1182 - 1144) - (-2220) = -2326 + 2220 = -106 \text{ kJ mol}^{-1}$.

EXAM TIP

The general shortcut used throughout the mark schemes is: **$\Delta_r H = \Sigma \Delta_f H(\text{products}) - \Sigma \Delta_f H(\text{reactants})$** when given formation data, or **$\Delta_r H = \Sigma \Delta_c H(\text{reactants}) - \Sigma \Delta_c H(\text{products})$** when given combustion data (note this one is the other way round). Get the direction wrong and every value in the calculation flips sign — if your answer has the wrong sign for an obviously exothermic reaction, check this first.

3.3 Bond enthalpy calculations

DEFINITION — Mean bond enthalpy

The **average** energy required to break one mole of a given bond in the gas phase, averaged across a range of different compounds containing that bond.

Bond-enthalpy calculations always use: **$\Delta_r H = \Sigma(\text{bonds broken in reactants}) - \Sigma(\text{bonds formed in products})$** . Breaking bonds is endothermic (energy in, positive); forming bonds is exothermic (energy out, subtract). Draw out the full structural/displayed formula first if the equation only gives molecular formulae — it is very easy to miscount how many of each bond type are present.

COMMON ERROR

Bond-enthalpy answers are always **less reliable** than data-book (formation/combustion) values, for two reasons you must be able to quote: (1) mean bond enthalpies are *averaged* over many different compounds, not measured for the specific molecule in the question; (2) bond enthalpy data refer to the **gaseous state** only, so using them for a reaction involving liquids or solids ignores the enthalpy changes of the state changes (vaporisation/melting) that must also occur.

3.4 Calorimetry calculations — $Q = mc\Delta T$

This is the single most repeated multi-mark calculation in the whole paper, appearing in some form on nearly every sitting. It always follows the same sequence, whether the context is combustion in a spirit burner, a displacement reaction in a polystyrene cup, or a dissolving/hydration enthalpy.

1. Calculate the energy change: **$Q = mc\Delta T$** (m = mass of the liquid being heated, in g — almost always water, with $c = 4.18 \text{ J g}^{-1} \text{ }^\circ\text{C}^{-1}$ unless the question states a different value, e.g. 4.2 or a value for a solution).
2. Convert Q to kJ if needed, and calculate the amount (mol) of the limiting/relevant substance that reacted or burned.
3. Divide: $\Delta H = -Q \div n$ (the negative sign appears because releasing heat to the surroundings, i.e. Q positive, means the reaction itself is exothermic, ΔH negative).
4. State the final answer to an appropriate number of significant figures (matching the least precise piece of data given) **with a sign and units (kJ mol^{-1})**.

WORKED EXAMPLE — Enthalpy of combustion from a spirit-burner experiment

0.320 g of an alcohol ($M_r = 60.0$) is burned and raises the temperature of 150 g of water by 12.4 °C. Specific heat capacity of water = 4.18 J g⁻¹ °C⁻¹.

$$Q = 150 \times 4.18 \times 12.4 = 7775 \text{ J} = 7.775 \text{ kJ}$$

$$n = 0.320 \div 60.0 = 5.33 \times 10^{-3} \text{ mol}$$

$$\Delta_c H = -7.775 \div (5.33 \times 10^{-3}) = -1460 \text{ kJ mol}^{-1} \text{ (3 s.f.)}$$

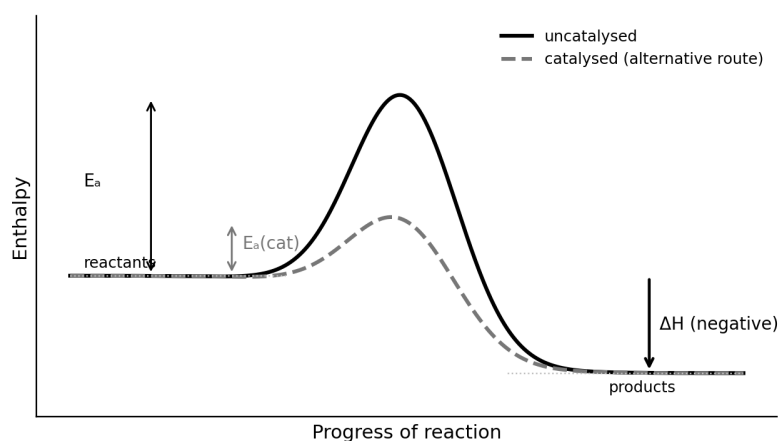
Sources of error in a calorimetry experiment — a very common short-answer question

When asked why an experimental value is *less exothermic* than the data-book value (other than “non-standard conditions”, which examiners usually explicitly exclude), the accepted reasons are:

- **Heat loss to the surroundings** — not all the heat released reaches the water/solution.
- **Incomplete combustion** — forms soot (carbon) or carbon monoxide instead of only CO₂, releasing less energy than complete combustion.
- **Evaporation of the fuel/alcohol** before or during burning, so not all of the mass weighed actually combusts.
- **Heat used to warm the apparatus itself** (the calorimeter/container), not accounted for in the calculation.

COMMON ERROR

Do **not** write “heat loss” as a catch-all for every question — if the question specifies “other than heat loss” or “other than non-standard conditions”, repeating that reason scores zero. Read exactly what has already been excluded.

3.5 Reaction profile diagrams and activation energy

A correctly labelled reaction profile: reactants and products at the correct relative heights, E_a measured from reactants to the peak (not from zero), ΔH measured from reactants to products, and the catalysed curve peaking lower via an alternative pathway while starting and finishing at exactly the same enthalpy levels as the uncatalysed curve.

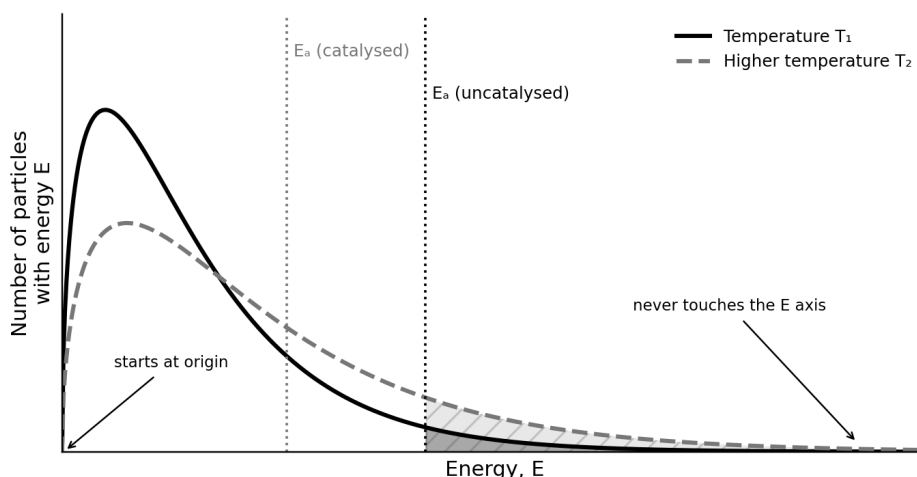
- **Exothermic** reactions: products drawn *lower* than reactants; ΔH is negative (an arrow pointing down from reactants to products).
- **Endothermic** reactions: products drawn *higher*; ΔH positive.
- **Activation energy, E_a** is always measured from the reactants line up to the peak of the curve — never from zero, and never from the products line.
- A catalyst lowers E_a by providing an **alternative reaction pathway/mechanism**. It does *not* change ΔH , and both curves must start and end at identical energy levels.

CHAPTER 4

Kinetics

Collision theory, Maxwell-Boltzmann, catalysts and rate from graphs

4.1 The Maxwell-Boltzmann distribution — draw it exactly right every time



The five features examiners check on every sketch/labelling question: starts at the origin, rises to a single peak, has a long right-hand tail that approaches but never touches the x-axis, encloses the same total area (same total number of particles) at any temperature, and shifts right/flattens (never simply “up”) as temperature increases.

- The curve **starts at the origin** (0,0) — no particles have zero energy.
- The curve **never touches the x-axis**, even at high energy — there is no maximum possible energy for a particle.
- The **area under the curve is constant** (equal to the total number of particles) regardless of temperature — a higher-temperature curve must therefore be *lower and broader* than a lower-temperature one, never simply the same shape shifted upward.
- As temperature increases: the peak moves to the right and gets lower/broader; the mean/most probable energy increases; and, crucially, a **greater proportion** of particles have energy $\geq E_a$.

■ DEFINITION — Why raising the temperature increases the rate — the exact chain of reasoning required

(1) Raising the temperature increases the average kinetic energy of the particles. (2) This means a greater proportion/number of particles now have energy $\geq$ the activation energy, E_a (shown as the increased area under the curve to the right of the E_a line). (3) This leads to a greater frequency of successful collisions per unit time. Every one of these three links is usually a separate marking point — do not skip straight from “particles move faster” to “rate increases”.

⚠ COMMON ERROR

“More collisions happen” on its own scores nothing — you must say **successful** collisions (i.e. collisions between particles with combined energy $\geq E_a$ and the correct orientation). Also do not say a catalyst “increases the energy of the particles” — a catalyst does not change the Maxwell-Boltzmann curve itself at all; it only lowers the activation energy needed, shifting the E_a line leftward on the same, unchanged curve.

4.2 How a catalyst increases rate

A catalyst provides an **alternative reaction pathway/mechanism with a lower activation energy**, without being used up in the overall reaction. On the Maxwell-Boltzmann diagram this is

shown by drawing the catalysed E_a line to the *left* of the uncatalysed E_a line, on the *same* curve — more particles now have energy $\geq$ the (lower) catalysed E_a , so the frequency of successful collisions increases.

A catalyst also appears repeatedly as a **sustainability** argument in 6-mark and short-answer questions. The accepted reasons a catalyst makes an industrial process “more sustainable” are:

- A catalyst allows a **lower operating temperature** to be used for an acceptable rate — this reduces energy costs (and associated CO₂ emissions from generating that energy).
- A specific/selective catalyst can reduce unwanted by-products, increasing atom economy.
- Because a catalyst is not consumed, it can be reused/recovered rather than needing constant replacement.

4.3 Reading rate from a graph

Concentration-time or volume-time graphs ask you to find the **initial rate** or the rate **at a specific time**. The method is identical in every appearance across the 18 papers studied:

1. Draw a **tangent** to the curve at the point specified (at $t = 0$ for an initial rate).
2. Choose two points on the tangent line that are easy to read accurately, as far apart as possible for accuracy.
3. Rate = gradient of the tangent = (change in y) $\div$ (change in x), e.g. $\Delta[\text{volume of gas}] \div \Delta[\text{time}]$.
4. Give the correct units — **cm³ s⁻¹** for a volume-time graph, or **mol dm⁻³ s⁻¹** (or per minute) for a concentration-time graph. Units must match whatever axis units were actually used.

♦ EXAM TIP

You must **show your tangent construction lines on the graph itself** — mark schemes explicitly require “working shown on the graph” for these marks, and a numerical answer with no visible tangent triangle can lose marks even if it is correct.

⚠ COMMON ERROR

When asked to sketch a second experiment on the same axes (e.g. larger solid lumps, lower concentration, added catalyst), think about **two separate features**: the *initial gradient* (steeper = faster rate) and the *final plateau height* (same amount of product if the same total moles of limiting reagent are used, unless the question changes the amount). A very common error is changing the wrong one — e.g. drawing a shallower curve that still finishes at a different height when the total moles of reactant have not changed.

CHAPTER 5

Equilibrium

Le Chatelier's Principle and choosing industrial conditions

5.1 Le Chatelier's Principle

■ DEFINITION — Le Chatelier's Principle

If a system at dynamic equilibrium is subjected to a change in concentration, pressure or temperature, the position of equilibrium shifts so as to **minimise/oppose** that change.

Change applied	Effect on position of equilibrium
Increase concentration of a reactant	Shifts right (towards products) to remove the added reactant
Increase pressure (gas equilibria)	Shifts towards the side with fewer moles of gas
Decrease pressure	Shifts towards the side with more moles of gas
Increase temperature	Shifts in the endothermic direction (absorbs the extra heat)
Decrease temperature	Shifts in the exothermic direction (releases heat to oppose the drop)
Catalyst added	No effect on position of equilibrium or yield — equilibrium is simply reached faster (both forward and reverse rates increase equally)

⚠ COMMON ERROR

Always distinguish **rate** from **position of equilibrium/yield** — these are marked as two separate ideas throughout the mark schemes. A change can increase the rate but decrease the yield at the same time (e.g. raising temperature for an exothermic reaction: rate always increases with temperature, but yield falls because equilibrium shifts left). Never assume “faster” automatically means “more product”.

5.2 Industrial conditions — the classic compromise argument

This exact style of reasoning (with different reactions each session: the Haber process, ethene + steam → ethanol, methanol synthesis, the Sabatier process, etc.) is one of the most frequently repeated 6-mark question types across the full set of papers — see Chapter 14 for the full mark-by-mark structure. The core logic to master now is:

- For an **exothermic** forward reaction: a **lower** temperature increases equilibrium yield (shifts right) but decreases rate; a **compromise, moderately high** temperature is therefore chosen to get an acceptable rate without losing too much yield.
- If there are fewer gas moles on the product side: **high pressure** increases both rate (particles more crowded, more frequent collisions) and yield (equilibrium shifts to the side with fewer moles) — but very high pressures are very expensive (stronger, more expensive equipment; higher running/energy costs), so a compromise pressure is used.
- A **catalyst** is used to increase the rate without needing to raise the temperature further — this is the “sustainability” link examiners want: a catalyst lets you keep the temperature (and therefore the energy cost, and the yield loss from an exothermic reaction) as low as possible while still reacting quickly enough.
- Any **unreacted gases are recycled** through the reactor, so even a low % yield per pass is acceptable and does not waste raw material overall.

CHAPTER 6

Redox, Oxidation Numbers & Disproportionation

The tool used inside almost every other chapter of this paper

6.1 Oxidation number rules

Appeared in every single one of the 18 papers studied, usually 2–4 separate times. Learn the hierarchy of rules — when two rules conflict, the rule higher up this list wins:

1. An uncombined element is always 0 (e.g. Cl_2 , O_2 , Na, S_8).
2. A simple monatomic ion has an oxidation number equal to its charge ($\text{Na}^+ = +1$, $\text{O}^{2-} = -2$).
3. The sum of oxidation numbers in a neutral compound = 0; in a polyatomic ion, the sum = the ion's overall charge.
4. Group 1 metals are always +1; Group 2 metals are always +2 in their compounds.
5. Fluorine is always -1 .
6. Hydrogen is +1 *except* in metal hydrides (e.g. NaH), where it is -1 .
7. Oxygen is -2 *except* in peroxides such as H_2O_2 (where it is -1) and in compounds with fluorine (e.g. OF_2 , where it is +2).

⚠ COMMON ERROR

A classic Section A trap is the oxidation number of oxygen in H_2O_2 (peroxide, -1 , not the usual -2) and in OF_2 or F_2O (+2, since fluorine outranks oxygen). Always check for peroxide or fluorine-containing exceptions before defaulting to -2 for oxygen.

6.2 Oxidising and reducing agents

- An **oxidising agent** is a species that oxidises something else — in doing so, it is itself **reduced** (gains electrons; its own oxidation number decreases).
- A **reducing agent** reduces something else — it is itself **oxidised** (loses electrons; its own oxidation number increases).
- Memory aid used throughout: **OIL RIG** — Oxidation Is Loss (of electrons), Reduction Is Gain.

✦ EXAM TIP

When comparing the reducing power of two halide ions (e.g. “which is the stronger reducing agent, bromide or iodide?”), the reasoning chain wanted is: the larger ion (further down the group) has its outer electrons further from the nucleus / more shielded, so an electron is lost more easily — therefore it is a better reducing agent, and this is confirmed by which halide is able to reduce concentrated sulfuric acid furthest (see §8.3).

6.3 Disproportionation

■ DEFINITION — Disproportionation

A redox reaction in which **the same element, in the same species, is simultaneously both oxidised and reduced** (i.e. one element ends up in two different oxidation states among the products).

The classic worked example, repeated across many sessions in different forms, is chlorine reacting with water or sodium hydroxide:

- Cold, dilute NaOH: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaOCl} + \text{H}_2\text{O}$ — chlorine goes from 0 in Cl_2 to -1 in Cl^- (NaCl, reduced) and to $+1$ in ClO^- (NaOCl, oxidised).

- Hot, concentrated NaOH: $\text{Cl}_2 + 6\text{NaOH} \rightarrow 5\text{NaCl} + \text{NaClO}_3 + 3\text{H}_2\text{O}$ — chlorine goes from 0 to -1 (reduced) and to $+5$ in ClO_3^- (oxidised).
- $\text{Cl}_2 + \text{H}_2\text{O} \rightleftharpoons \text{HOCl} + \text{HCl}$ — chlorine goes from 0 to $+1$ (in HOCl) and to -1 (in HCl).

⚠ COMMON ERROR

To fully explain “why is this disproportionation” you must state the actual **oxidation number before and after** for both the oxidised and the reduced product — simply writing “chlorine changes oxidation state” without the numbers scores less than full marks. Also learn to recognise the reverse question: given only the oxidation numbers, decide whether a reaction is disproportionation, simple redox, or not redox at all (no oxidation number changes).

6.4 Half-equations — writing and combining them

- Balance the atoms undergoing oxidation/reduction first.
- Balance oxygens by adding H_2O .
- Balance hydrogens by adding H^+ .
- Balance charge by adding electrons (e^-) to the more positive side.

To **combine two half-equations** into an overall ionic equation, scale each one so the number of electrons is equal on both, then add them together and cancel the electrons (and anything else appearing identically on both sides).

🔗 WORKED EXAMPLE — Combining half-equations

Given: $2\text{I}^- \rightarrow \text{I}_2 + 2\text{e}^-$ and $8\text{H}^+ + \text{H}_2\text{SO}_4 + 8\text{e}^- \rightarrow \text{H}_2\text{S} + 4\text{H}_2\text{O}$.

Scale the iodide equation $\times 4$ so electrons match (8e^- both sides): $8\text{I}^- \rightarrow 4\text{I}_2 + 8\text{e}^-$.

Add and cancel the 8e^- : **$8\text{I}^- + 8\text{H}^+ + \text{H}_2\text{SO}_4 \rightarrow 4\text{I}_2 + \text{H}_2\text{S} + 4\text{H}_2\text{O}$** .

CHAPTER 7

Group 1 & Group 2

Reactivity trends, flame tests, solubility and thermal decomposition

7.1 Reactivity trends down the group

Both groups become **more reactive down the group** because the atoms get larger (more electron shells), so the outer electron(s) are further from the nucleus and more shielded from the nuclear charge by inner shells — this reduces the attraction holding the outer electron(s) in place, so ionisation energy **decreases** down the group and the electron(s) are lost more easily.

Observation when the metal reacts with water	Group 1 (e.g. Na, K)	Group 2 (e.g. Ca, Sr, Ba)
Vigour	Increasingly vigorous down the group; K catches fire (lilac flame)	Less vigorous than Group 1; increases down the group (Mg reacts very slowly with cold water, but readily with steam)
Products	Metal hydroxide (soluble) + H ₂ gas	Metal hydroxide + H ₂ gas
Typical equation	$2\text{Na(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)} + \text{H}_2\text{(g)}$	$\text{Ca(s)} + 2\text{H}_2\text{O(l)} \rightarrow \text{Ca(OH)}_2\text{(aq)} + \text{H}_2\text{(g)}$

7.2 Flame test colours — learn this table exactly

Ion	Flame colour
Li ⁺	Red (crimson)
Na ⁺	Yellow/orange
K ⁺	Lilac
Rb ⁺	Red
Cs ⁺	Blue/violet
Ca ²⁺	Brick-red / orange-red
Sr ²⁺	Red (crimson)
Ba ²⁺	Green (apple/pale green)

■ DEFINITION — How a flame colour forms — full mark-scheme wording

Heat energy **excites/promotes electrons** to a higher energy level. These electrons then **fall back down to the ground state (or a lower energy level)**, and in doing so **release energy as light of a specific wavelength/frequency**, which corresponds to a particular visible colour.

⚠ COMMON ERROR

All three steps above (excitation, falling back, release of light/energy of a specific wavelength) are usually separate marking points — “electrons emit a colour” on its own, without mentioning excitation and falling back, typically scores only 1 of 3 marks.

7.3 Solubility trends

- Group 2 **hydroxides** become **more soluble** down the group (Mg(OH)₂ is barely soluble; Ba(OH)₂ is fairly soluble — this is why limewater, a saturated Ca(OH)₂ solution, is only weakly alkaline).

- Group 2 **sulfates** become **less soluble** down the group (MgSO_4 is very soluble; BaSO_4 is essentially insoluble — exploited in the qualitative test for sulfate ions using acidified BaCl_2 , and in the test to distinguish Group 2 ions from each other).

7.4 Thermal stability of nitrates and carbonates — a guaranteed topic (17/18 papers)

Group 2 carbonates and nitrates become **more thermally stable down the group** (need a higher temperature to decompose). The full explanation, built from several separate marking points, is:

- Going down the group, the metal cation gets **larger** (increasing ionic radius).
- A larger cation has a **lower charge density** and is therefore **less polarising** — it distorts the electron cloud of the neighbouring carbonate/nitrate anion less.
- This means the covalent bond inside the anion (the C-O bond in CO_3^{2-} , or the N-O bond in NO_3^-) is **weakened less**, so a higher temperature is needed to break it and cause decomposition.

⚠ COMMON ERROR

The polarisation is the **cation polarising the anion**, not the other way round — and the bond that breaks/weakens is the **covalent bond inside the carbonate/nitrate ion** (e.g. C-O), never described as an “ionic bond breaking”. This exact error (“the anion polarises the cation”, or “the ionic bond weakens”) is explicitly penalised in several mark schemes — double-check the direction every time you write this answer.

Decomposition type	Typical equation	Pattern
Group 2 carbonate	$\text{MgCO}_3(\text{s}) \rightarrow \text{MgO}(\text{s}) + \text{CO}_2(\text{g})$	All Group 2 carbonates decompose (on Bunsen heating) to the metal oxide + CO_2 ; stability increases $\text{Mg} < \text{Ca} < \text{Sr} < \text{Ba}$
Group 2 nitrate	$2\text{Mg}(\text{NO}_3)_2(\text{s}) \rightarrow 2\text{MgO}(\text{s}) + 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$	All Group 2 nitrates decompose to the metal oxide + brown NO_2 gas + O_2
Group 1 nitrate (except Li)	$2\text{NaNO}_3(\text{s}) \rightarrow 2\text{NaNO}_2(\text{s}) + \text{O}_2(\text{g})$	Group 1 nitrates (Na, K, Rb, Cs) decompose only as far as the metal nitrite + O_2 — no brown NO_2 gas is seen
Lithium nitrate (the exception)	$4\text{LiNO}_3(\text{s}) \rightarrow 2\text{Li}_2\text{O}(\text{s}) + 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$	Li^+ behaves like a Group 2 ion (very small, high charge density) and decomposes like a Group 2 nitrate, not like the rest of Group 1

♦ EXAM TIP

“Why does lithium nitrate behave like Group 2, not Group 1?” has appeared as an extended (6-mark) question. The link is: Li^+ is unusually small for Group 1 (similar ionic radius to Mg^{2+}), so it has an unusually high charge density and polarises the nitrate ion enough to force full decomposition to the oxide + NO_2 , in the same way Group 2 cations do.

CHAPTER 8

Group 7 — The Halogens

Trends, displacement, halide tests and chlorine chemistry

8.1 Trends down the group

Property	Trend down the group (F → Cl → Br → I)	Reason
Electronegativity	Decreases	Atomic radius increases (more shells), so the nuclear attraction for a bonding pair of electrons is weaker, despite the increasing nuclear charge
Boiling temperature	Increases (F ₂ , Cl ₂ gases; Br ₂ liquid; I ₂ solid)	More electrons per molecule → stronger London (dispersion) forces between molecules, needing more energy to overcome
Oxidising power	Decreases	Larger atoms attract an incoming electron less strongly (electron added further from, and more shielded from, the nucleus), so are weaker oxidising agents
Reactivity (as an oxidising agent)	Decreases	Follows directly from the decrease in oxidising power

8.2 Displacement reactions and colours

A more reactive (more oxidising) halogen displaces a less reactive halide from solution, e.g. Cl₂(aq) + 2Br⁻(aq) → 2Cl⁻(aq) + Br₂(aq). Colour observations are frequently tested, both in aqueous solution and after shaking with a non-polar solvent (hexane or cyclohexane), which the halogen product dissolves into and concentrates in, making the colour easier to see:

Halogen	Colour in aqueous solution	Colour in hexane / cyclohexane (upper, non-polar layer)
Chlorine, Cl ₂	Pale green / colourless	Very pale green / colourless
Bromine, Br ₂	Orange	Orange
Iodine, I ₂	Brown	Purple / violet

⚠ COMMON ERROR

Iodine's colour change (brown in water → purple/violet in a non-polar solvent) is the one most often tested, because it is the only halogen that changes colour on going from the aqueous to the organic layer — a very common Section A distractor is to give iodine the same colour in both layers.

8.3 Reactions of solid halides with concentrated sulfuric acid — learn this whole table

This reaction sequence is one of the most heavily tested redox contexts in the whole specification. All four halides first undergo simple **acid-base (proton transfer)** reactions, releasing steamy/misty fumes of the hydrogen halide — **this first step is not redox**. Only bromide and iodide then go on to reduce the sulfuric acid further, because only they are strong enough reducing agents:

Halide	Step 1 (always, non-redox)	Further redox reaction(s)	Observations
Chloride, Cl ⁻	NaCl + H ₂ SO ₄ → NaHSO ₄ + HCl	None — Cl ⁻ cannot reduce H ₂ SO ₄	Misty/steamy white fumes only

Halide	Step 1 (always, non-redox)	Further redox reaction(s)	Observations
Bromide, Br ⁻	$\text{NaBr} + \text{H}_2\text{SO}_4 \rightarrow \text{NaHSO}_4 + \text{HBr}$	$2\text{HBr} + \text{H}_2\text{SO}_4 \rightarrow \text{Br}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$ (S: +6 → +4)	Misty fumes, plus orange/brown vapour/liquid (Br ₂) and choking/pungent SO ₂ gas
Iodide, I ⁻	$\text{NaI} + \text{H}_2\text{SO}_4 \rightarrow \text{NaHSO}_4 + \text{HI}$	$6\text{HI} + \text{H}_2\text{SO}_4 \rightarrow 3\text{I}_2 + \text{S} + 4\text{H}_2\text{O}$ and/or $8\text{HI} + \text{H}_2\text{SO}_4 \rightarrow 4\text{I}_2 + \text{H}_2\text{S} + 4\text{H}_2\text{O}$ (S: +6 → 0 or -2)	Misty fumes, purple vapour/black solid (I ₂), plus yellow solid (S) and/or a 'bad eggs' smell (H ₂ S)

✦ EXAM TIP

The reasoning link required: **iodide is a stronger reducing agent than bromide** because I⁻ is a larger ion, so its outer electron is lost more easily — this is why iodide reduces sulfur all the way to -2 (H₂S) while bromide only manages +4 (SO₂), and chloride cannot reduce sulfur at all.

8.4 Testing for halide ions with silver nitrate

Aqueous silver nitrate is added (usually after acidifying with dilute nitric acid, to remove interference from other anions such as carbonate or hydroxide). The precipitate colour identifies the halide, and its solubility in ammonia confirms it further:

Halide	Precipitate colour	Solubility in ammonia
Cl ⁻	White	Dissolves in dilute (or concentrated) aqueous ammonia
Br ⁻	Cream / off-white	Insoluble in dilute ammonia; dissolves in concentrated ammonia
I ⁻	Yellow	Insoluble even in concentrated ammonia

Ionic equation (learn generally, for any halide X⁻): $\text{Ag}^+(\text{aq}) + \text{X}^-(\text{aq}) \rightarrow \text{AgX}(\text{s})$

8.5 Chlorine with water and with sodium hydroxide

- **Cold water:** $\text{Cl}_2(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{HOCl}(\text{aq}) + \text{HCl}(\text{aq})$ — an equilibrium, and a disproportionation (0 → +1 and 0 → -1).
- **Cold, dilute NaOH** (household bleach conditions): $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaOCl} + \text{H}_2\text{O}$.
- **Hot, concentrated NaOH:** $3\text{Cl}_2 + 6\text{NaOH} \rightarrow 5\text{NaCl} + \text{NaClO}_3 + 3\text{H}_2\text{O}$.

⚠ COMMON ERROR

If asked why mixing bleach with a strong (household) acid is dangerous: acid shifts the equilibria $\text{HClO}(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{ClO}^-(\text{aq})$ and $\text{HClO} + \text{Cl}^- + \text{H}^+ \rightleftharpoons \text{Cl}_2 + \text{H}_2\text{O}$ to the **right** (removing H⁺ as it is used up drives more HClO to form, and pushes the second equilibrium towards Cl₂), releasing toxic chlorine gas — you must reference the actual equilibria, not just state “it's dangerous”.

CHAPTER 9

Intermolecular Forces & Physical Properties

Explaining boiling temperature, solubility and state — tested in every paper

9.1 The three intermolecular forces, ranked by strength

Force	Present between	Origin	Relative strength
London (dispersion) forces	All molecules and atoms, polar or non-polar (including noble gases)	A temporary/instantaneous dipole (from momentarily uneven electron distribution) induces a dipole in a neighbouring molecule	Weakest — but the only force present between totally non-polar molecules, so still decisive for e.g. the halogens or alkanes
Permanent dipole-permanent dipole	Polar molecules only (unsymmetrical, containing atoms of different electronegativity)	Permanent $\delta+$ and $\delta-$ charges on neighbouring molecules attract	Stronger than London forces between molecules of similar size
Hydrogen bonding	Only where H is bonded directly to N, O or F (the three most electronegative, small elements)	A very strong permanent dipole between the $\delta+$ H and a lone pair on N/O/F of a neighbouring molecule	Strongest intermolecular force — far stronger than the other two

■ DEFINITION — Criteria for hydrogen bonding — all three must be true

(1) A hydrogen atom bonded directly to N, O or F; (2) that N/O/F atom has a lone pair of electrons; (3) the H-(N/O/F) bond is highly polar because N/O/F is small and very electronegative. All molecules containing -OH, -NH or -NH₂ groups (water, alcohols, ammonia, amines, carboxylic acids) can hydrogen-bond; ketones, aldehydes, ethers and esters **cannot** (no H directly bonded to O), even though they contain oxygen.

⚠ COMMON ERROR

A very common Section A/B trap: HF has hydrogen bonding but boils at a much lower temperature than water despite both hydrogen-bonding, because **each water molecule can form twice as many hydrogen bonds** (2 lone pairs and 2 O-H bonds vs. HF's 3 lone pairs but only 1 H atom, effectively limited to 1 hydrogen bond per molecule on average). Always compare the *number* of hydrogen bonds per molecule, not just whether hydrogen bonding is present.

9.2 Explaining boiling temperature trends — the exact structure examiners want

This appears, in some form, on **every single paper** studied — sometimes as a 1-mark MCQ, sometimes as a 6-mark extended question comparing 3–4 substances. The required structure is always the same:

1. Identify **which intermolecular force(s)** are present in each substance (all substances have London forces; add permanent dipole and/or hydrogen bonding as applicable).
2. State that **London forces increase with more electrons** (bigger molecule/more atoms) — use this to compare boiling temperature *within* a series where the same type(s) of force are present.
3. State that **hydrogen bonding is much stronger than the other forces**, so any substance capable of hydrogen bonding will usually have an unusually high boiling temperature, even if it has fewer electrons than a rival substance with only London/permanent-dipole forces.

4. Conclude: more/stronger intermolecular forces → **more energy is required to overcome them** → higher boiling temperature.

 WORKED EXAMPLE — Fully worked comparison

Explain why methanol ($T_b = 65\text{ }^{\circ}\text{C}$) has a much higher boiling temperature than fluoromethane ($T_b = -78\text{ }^{\circ}\text{C}$), despite similar Mr.

Both molecules have London forces of similar strength (similar number of electrons). Fluoromethane also has permanent dipole-permanent dipole forces (C-F is polar) but **no hydrogen bonding**, because although F is present, no H atom is bonded directly to it. Methanol has an O-H bond, so it **can** hydrogen bond between molecules. Hydrogen bonds are much stronger than permanent dipole-permanent dipole forces, so significantly more energy is needed to separate methanol molecules — giving it the higher boiling temperature.

9.3 Solubility — “like dissolves like”

- A solute dissolves well in a solvent when the **solute-solvent** intermolecular forces that can form are of **similar strength** to the forces being broken (solute-solute and solvent-solvent).
- Ionic compounds and small polar/hydrogen-bonding molecules (e.g. short alcohols) dissolve well in water because water molecules can hydrogen-bond to them (or, for ions, form strong ion-dipole attractions by orienting δ^- oxygen towards cations and δ^+ hydrogens towards anions).
- Non-polar substances (e.g. iodine, long-chain hydrocarbons) dissolve poorly in water (only weak forces could form, far weaker than the hydrogen bonds within water that would need breaking) but dissolve well in non-polar solvents such as hexane or cyclohexane (only London forces need to form, matching the London forces already present in both).
- Long-chain alcohols become progressively **less soluble** in water as the hydrocarbon chain lengthens, because the non-polar alkyl part becomes an increasingly large proportion of the molecule, while only the single -OH group can hydrogen bond.

CHAPTER 10

Halogenoalkanes

Nucleophilic substitution, elimination and hydrolysis rates — in 18/18 papers

10.1 Classification

A halogenoalkane is classified by **how many carbon (alkyl) groups are attached to the carbon bearing the halogen**: **primary** (1 alkyl group, e.g. $\text{CH}_3\text{CH}_2\text{Br}$), **secondary** (2 alkyl groups, e.g. $(\text{CH}_3)_2\text{CHBr}$) or **tertiary** (3 alkyl groups, e.g. $(\text{CH}_3)_3\text{CBr}$). This classification determines which substitution mechanism operates — see below.

10.2 Nucleophilic substitution — the two mechanisms

A halogenoalkane's C-X bond is polarised ($\text{C } \delta+$, $\text{X } \delta-$, since halogens are more electronegative than carbon), making the carbon susceptible to attack by an electron-rich **nucleophile** (a lone-pair or negatively-charged species). You must be able to draw both mechanisms completely, with correct curly arrows, dipoles, lone pairs and any charges.

SN2 (primary halogenoalkanes) — one step

- The nucleophile (e.g. OH^-) attacks the $\delta+$ carbon from the **side opposite** the leaving halogen (back-side attack), while the C-X bond breaks simultaneously.
- One curly arrow: from the nucleophile's lone pair to the carbon atom. A second curly arrow: from the C-X bond to the halogen (or just beyond it, onto the halogen symbol) to show X^- leaving.
- Single step, no charged intermediate carbon — this is why primary halogenoalkanes react cleanly and predictably.

SN1 (tertiary halogenoalkanes) — two steps via a carbocation

- Step 1**: the C-X bond breaks heterolytically *first*, without any nucleophile involved yet, forming a planar, positively charged **tertiary carbocation intermediate** and a free halide ion. One curly arrow: from the C-X bond to the halogen.
- Step 2**: the nucleophile's lone pair then attacks the positively charged carbon of the carbocation. One curly arrow: from the nucleophile's lone pair to the carbocation carbon.
- The carbocation intermediate is **trigonal planar** (the charged carbon is sp^2 hybridised, three bonds only) — if asked to justify the shape, state this explicitly.

⚠ COMMON ERROR

Secondary halogenoalkanes can react by **either** mechanism (a mixture), which is why exam questions often say “it is not possible to predict” the exact rate/mechanism for a secondary halogenoalkane with confidence. Common mechanism-drawing errors that lose marks: missing a lone pair on the nucleophile, a single-headed (‘fish-hook’) arrow instead of a full double-headed curly arrow, or an arrow starting from the atom itself rather than from a bond/lone pair.

10.3 Rate of hydrolysis — why iodoalkanes react fastest

When a series of halogenoalkanes (e.g. 1-chlorobutane, 1-bromobutane, 1-iodobutane) are warmed with aqueous silver nitrate in ethanol, a precipitate of the silver halide forms — the faster the precipitate appears, the faster that C-X bond hydrolyses. The rate order is always **iodo > bromo > chloro**, and the explanation is entirely about **bond enthalpy, not electronegativity**:

■ DEFINITION — Why C-I hydrolyses fastest

Going down Group 7, atomic radius increases, so the carbon-halogen bond gets **longer**. A longer bond is **weaker** (lower bond enthalpy) and therefore breaks more easily/faster — giving C-I the fastest rate of hydrolysis and C-Cl the slowest, even though C-Cl is the most polar of the three bonds.

⚠ COMMON ERROR

This is one of the most consistently repeated “do not award” traps across every mark scheme studied: do **not** explain the rate trend using electronegativity or bond polarity — C-Cl is the *most* polar bond of the three, yet reacts *slowest*, so polarity cannot be the deciding factor. The explanation must be about **bond strength/enthalpy** only.

Separately, **tertiary** halogenoalkanes hydrolyse faster than primary ones with the *same* halogen, because the SN1 mechanism (via a relatively stable tertiary carbocation) has a lower activation energy than SN2 — if you are given a comparison of, say, 2-bromo-2-methylpropane vs. 1-bromobutane, this is the reasoning wanted, not bond enthalpy (the C-Br bond enthalpy is the same in both).

10.4 Elimination — forming alkenes

Halogenoalkanes heated under reflux with **potassium hydroxide dissolved in ethanol** (never aqueous) undergo elimination instead of substitution. Here the hydroxide ion acts as a **base**, not a nucleophile — it removes a H⁺ from a carbon *adjacent* to the C-X carbon, and a C=C double bond forms as the halide ion leaves.

- Curly arrow 1: from a lone pair on OH⁻ to the adjacent (β) hydrogen.
- Curly arrow 2: from the C-H bond into the space between the two carbons, forming the new C=C π bond.
- Curly arrow 3: from the C-X bond onto the halogen, as it leaves as X⁻.
- Product: H₂O + KX + the alkene.

⚠ COMMON ERROR

The **solvent** is the deciding factor between substitution and elimination with the same reagent (KOH): **aqueous** KOH → substitution (OH⁻ acts as nucleophile, alcohol forms); **ethanolic** KOH, heated → elimination (OH⁻ acts as base, alkene forms). If a halogenoalkane has hydrogens on more than one adjacent carbon, more than one alkene (including E/Z isomers of one structural isomer) can be formed — always check which β-carbons carry a removable H.

10.5 Other important reactions of halogenoalkanes

Product wanted	Reagent	Conditions	Notes
Alcohol	KOH (or NaOH)	Aqueous, heat under reflux	Nucleophilic substitution; OH ⁻ is the nucleophile
Nitrile (chain extension)	KCN	In ethanol, heat under reflux	Adds one extra carbon to the chain — the nitrile can then be reduced (H ₂ /Pt or LiAlH ₄) to a primary amine, or hydrolysed to a carboxylic acid
Primary amine	Excess concentrated NH ₃	In ethanol, heated in a sealed tube (under pressure) — not simple reflux, since ammonia would escape	Excess ammonia is used to minimise further substitution (secondary/tertiary amine formation, since the product amine is itself a nucleophile)

Product wanted	Reagent	Conditions	Notes
Alkene	KOH	Ethanolic, heat under reflux	Elimination — see §10.4
♦ EXAM TIP			
“Why is the nitrile route important in synthesis?” — the accepted answer is simply that it is a way of extending/lengthening the carbon chain by one carbon atom , which substitution with OH [−] or NH ₃ alone cannot do.			

CHAPTER 11

Alcohols

Classification, oxidation and conversion to halogenoalkanes

11.1 Classification

Exactly as for halogenoalkanes: **primary** alcohols have the C-OH carbon attached to one alkyl group (and 2 H atoms), **secondary** to two alkyl groups (and 1 H), **tertiary** to three alkyl groups (no H on that carbon). This classification controls what oxidation is possible — see below.

11.2 Oxidation with acidified potassium dichromate(VI)

The oxidising mixture is **potassium dichromate(VI)**, $K_2Cr_2O_7$, **acidified with dilute sulfuric acid**. The colour change on a successful oxidation is always **orange** → **green** ($Cr_2O_7^{2-}$, orange, +6, is reduced to Cr^{3+} , green, +3).

Alcohol type	Product & apparatus	Why
Primary	Distil off as it forms → aldehyde only. Heat under reflux (excess oxidant) → carboxylic acid	Distillation removes the aldehyde before it can be oxidised further; reflux keeps the oxidant in contact with the product so oxidation continues to completion
Secondary	Heat under reflux → ketone (no further oxidation possible)	The carbon has no second H atom to lose, so oxidation stops at the ketone
Tertiary	No reaction (resists oxidation by acidified dichromate)	The carbon bonded to -OH has no H atom attached at all, so it cannot be oxidised by this method

♦ **EXAM TIP — Distinguishing the products afterwards — always link two techniques together**

- **Chemical test:** Fehling's or Benedict's solution (blue Cu^{2+} solution) — an aldehyde gives a colour change to a **brick-red precipitate** (Cu_2O); a ketone gives **no change** (remains blue), since it cannot be oxidised further. Acidified potassium dichromate itself can also be used as the distinguishing test (orange → green with an aldehyde; no colour change with a ketone).
- **Infrared spectroscopy:** as oxidation proceeds, the broad O-H absorption ($3750-3200\text{ cm}^{-1}$, alcohol) diminishes/disappears and a new, sharp C=O absorption ($1740-1700\text{ cm}^{-1}$) appears. If oxidation continues to a carboxylic acid, the O-H peak reappears but now very broad ($3300-2500\text{ cm}^{-1}$) due to hydrogen bonding between acid molecules, overlapping the C-H region.

⚠ **COMMON ERROR**

“Do not award reflux” appears repeatedly for the *aldehyde*-forming step, and “do not award distillation” for the carboxylic-acid step — these two apparatus choices are opposite, and mixing them up (or just saying “heat” without specifying which) loses the mark even if the reagent named is correct.

11.3 Converting alcohols into halogenoalkanes

Halogen wanted	Reagent	Observation used as a chemical test for -OH	Notes
Chloro-	PCl_5 (solid), or conc. HCl (+ $ZnCl_2$ catalyst, for the Lucas test), or $SOCl_2$	PCl_5 gives misty/steamy white fumes of HCl on contact — the standard test for an -OH group	Room temperature is usually enough for PCl_5
Bromo-	$NaBr/KBr$ solid + concentrated H_2SO_4 (generates HBr <i>in situ</i>), or PBr_3	Misty fumes, as above	Heat under reflux

Halogen wanted	Reagent	Observation used as a chemical test for -OH	Notes
Iodo-	Red phosphorus + I ₂ (generates PI ₃ <i>in situ</i>)	—	Heat under reflux

PCl₅ with an alcohol is also the standard **chemical test to confirm the presence of an -OH group**: $\text{ROH} + \text{PCl}_5 \rightarrow \text{RCl} + \text{POCl}_3 + \text{HCl}$, observed as immediate misty/steamy white fumes of HCl (which themselves form a white smoke with concentrated ammonia, confirming HCl specifically).

11.4 Dehydration of alcohols to alkenes

Heating an alcohol with a concentrated acid catalyst (**concentrated phosphoric(V) acid, H₃PO₄, or concentrated sulfuric acid**) causes **elimination** of water, forming an alkene. This is the reverse of the industrial hydration of ethene to make ethanol ($\text{CH}_2=\text{CH}_2 + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{CH}_2\text{OH}$, catalysed by H₃PO₄ at 300 °C, 60 atm — a classic 6-mark ‘choice of conditions’ context, see Chapter 14).

⚠ COMMON ERROR

If the alcohol can lose an H from more than one side of the C-OH carbon, more than one alkene product is possible (including structural and, separately, E/Z geometric isomers) — always check every β-carbon with an available H, exactly as for halogenoalkane elimination.

11.5 Physical properties of alcohols

- Alcohols can hydrogen-bond (via the -OH group), giving them noticeably higher boiling temperatures than alkanes or halogenoalkanes of similar Mr, and giving short-chain alcohols (methanol, ethanol, propanols) full miscibility with water.
- Solubility in water **decreases** as the carbon chain lengthens, because the non-polar hydrocarbon portion (which cannot hydrogen bond, and disrupts the hydrogen-bonded structure of water) makes up a larger fraction of the molecule.
- Branching generally **lowers** boiling temperature for a given number of carbons (less surface contact between molecules, weaker London forces) — the same rule as for alkanes and halogenoalkanes.

CHAPTER 12

Spectroscopy — IR & Mass Spec

Identifying and distinguishing organic compounds — in 16/18 and 11/18 papers

12.1 Infrared spectroscopy

Every IR question expects you to quote a **wavenumber range** (from the Data Booklet) alongside the bond responsible — a bond name with no range, or a range with no bond, typically only earns partial credit. The functional groups that repeatedly appear across all 18 papers are:

Bond	Approx. wavenumber / cm^{-1}	Found in
O-H (alcohol)	3750–3200 (broad)	Alcohols
O-H (carboxylic acid)	3300–2500 (very broad, often overlapping C-H)	Carboxylic acids
N-H	3500–3300	Amines, amides
C-H	3100–2850	Almost all organic compounds — rarely diagnostic on its own
$\text{C}\equiv\text{N}$ (nitrile)	2260–2220	Nitriles
$\text{C}=\text{O}$	1740–1700	Aldehydes, ketones, carboxylic acids, esters
$\text{C}=\text{C}$	1669–1645 (often weak)	Alkenes

♦ **EXAM TIP** — The standard “distinguish two compounds using IR” answer template

State which peak is **present in one spectrum but absent in the other** (or vice versa), including both the correct wavenumber range *and* which bond it corresponds to, for **each** compound being compared. E.g. “the alcohol shows a broad O-H peak at 3750–3200 cm^{-1} which is absent in the aldehyde; the aldehyde instead shows a $\text{C}=\text{O}$ peak at 1740–1700 cm^{-1} which is absent in the alcohol.” Simply saying “the spectra are different” scores nothing.

⚠ **COMMON ERROR**

Never base an answer on the **fingerprint region** (below $\sim 1500 \text{ cm}^{-1}$) unless specifically permitted — several mark schemes explicitly instruct “ignore comments on the fingerprint region” except where the question allows it. Also always ignore C-H absorptions when comparing two compounds unless they are the only difference, since C-H is present in virtually everything.

12.2 Mass spectrometry

- The **molecular ion peak, M^+** , is always the peak with the **highest m/z value** in the spectrum (occasionally excluding very small isotope peaks) — its m/z value equals the relative molecular mass, M_r , of the compound.
- Fragmentation**: the molecular ion breaks apart in the spectrometer; each fragment peak corresponds to a positively charged piece of the original molecule. You are expected to assign a plausible formula/structure to a given m/z fragment peak.
- Common diagnostic fragments: **$m/z = 15$** (CH_3^+), **$m/z = 29$** (CHO^+ or C_2H_5^+), **$m/z = 43$** (CH_3CO^+ , the acylium ion — very characteristic of methyl ketones, or C_3H_7^+), **$m/z = 45$** (COOH^+ or CH_3CHOH^+).

WORKED EXAMPLE — Typical reasoning for identifying a compound from Mr

A compound has M^+ at $m/z = 74$. If it is known to contain C, H and one O atom, calculate a molecular formula consistent with 74 (e.g. $C_4H_{10}O$, $M_r = 74$) and use any chemical test results (e.g. does it react with 2,4-DNP, does it effervesce with Na_2CO_3 , does it decolourise bromine water) to decide between the possible isomers.

12.3 'Identify the unknown' combined questions

Several papers give both an IR and a mass spectrum (sometimes plus chemical test results) for an unknown, together with 2–3 possible candidate structures, and ask you to identify which one it is. Always work systematically:

1. Read off the **M^+ value** from the mass spectrum and eliminate any candidate whose M_r does not match.
2. Check the **IR spectrum** for the presence/absence of key functional-group peaks (O–H, C=O, C=C, etc.) and eliminate any candidate that is inconsistent.
3. If a **base peak** (the tallest peak, most abundant fragment) is given, check it matches a sensible fragment of the remaining candidate(s).
4. State your final answer explicitly, and justify it by referencing *all* the pieces of evidence you used, not just one.

CHAPTER 13

Practical & Quantitative Skills

Titration, moles, gas calculations, atom economy and errors

13.1 The core moles toolkit

Quantity	Formula
Moles from mass	$n = \text{mass (g)} \div M_r$
Moles from concentration & volume	$n = \text{concentration (mol dm}^{-3}\text{)} \times \text{volume (dm}^3\text{)}$ [volume in $\text{cm}^3 \div 1000$]
Moles of gas (at r.t.p.)	$n = \text{volume (cm}^3\text{)} \div 24\,000$ (or $\div 24.0$ if volume is in dm^3 ; r.t.p. molar volume is given in the question, commonly $24\,000 \text{ cm}^3 \text{ mol}^{-1}$ or $24.0 \text{ dm}^3 \text{ mol}^{-1}$)
Moles of gas (any conditions)	$pV = nRT$, so $n = pV \div RT$ (p in Pa, V in m^3 , T in K, $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)
Percentage yield	$(\text{actual moles/mass obtained} \div \text{theoretical maximum moles/mass}) \times 100$
Percentage atom economy	$(M_r \text{ of desired product} \div \text{sum of } M_r \text{ of all products}) \times 100$, using the balanced equation
⚠ COMMON ERROR	
For $pV = nRT$ questions: temperature must be converted to kelvin (+273), pressure is usually given in kPa and must be converted to Pa ($\times 1000$), and volume is usually given in cm^3 and must be converted to m^3 ($\div 10^6$) — a very common source of lost marks is forgetting one of these three unit conversions, even when the rest of the method is correct.	

13.2 Titration calculations

1. Calculate moles of the substance with known concentration, using $n = c \times V$.
2. Use the **balanced equation's mole ratio** to find moles of the other substance.
3. Convert to whatever the question asks for (concentration, mass, M_r , percentage purity, etc.), remembering to scale up correctly if only a fraction of a larger prepared solution was titrated (e.g. multiply by 10 if only a 25.0 cm^3 sample out of a 250.0 cm^3 volumetric flask was used).

✎ WORKED EXAMPLE — Standard 'find x in a hydrate/find purity' style titration

A 1.00 g sample of an impure Group 2 carbonate is reacted with a known excess of standardised HCl; the leftover (unreacted) acid is then titrated with standardised NaOH.

- moles HCl added (known excess) – moles NaOH used in titration (scaled to the whole flask) = moles HCl that actually reacted with the carbonate.
- Use the equation's ratio (usually 1 : 2, carbonate : HCl) to find moles of carbonate, then $M_r = \text{mass} \div \text{moles}$, and compare to the known M_r of CO_3^{2-} (60) to identify the metal, or compare to the sample mass for % purity.

Practical technique points that are regularly tested as short-answer questions:

- A pipette is used to measure a fixed, accurate volume for transfer into the conical flask; a burette delivers the variable, measured volume (the titre).
- **Concordant results** are titres that agree within 0.10 cm^3 of each other — only concordant titres should be averaged (discard any 'rough' or outlying titre first).

- **Percentage uncertainty** in a single burette reading is usually taken as $\pm 0.05 \text{ cm}^3$; since a titre is found by subtracting two readings, the uncertainty in the titre itself is **$\pm 0.10 \text{ cm}^3$** (twice the single-reading uncertainty). Percentage uncertainty = (uncertainty $\div$ measured value) $\times 100$.
- Rinsing glassware: a burette/pipette is rinsed with the solution it will contain (not with water, and not with any other reagent); a conical flask is rinsed only with distilled water (rinsing it with the reagent would add extra moles and give a systematic error).

13.3 Empirical and molecular formula

1. Convert each given mass (or percentage by mass) to moles by dividing by that element's A_r .
2. Divide every mole value by the smallest one, to get a ratio.
3. Round to the nearest whole-number ratio (multiply all values up if necessary to clear an obvious fraction, e.g. $\times 2$ if you see a consistent .5).
4. For the **molecular** formula, compare the empirical formula mass to the M_r (often from a mass spectrum's M^+ peak) and scale up by the appropriate whole-number multiple.

13.4 Significant figures, sign and units — do not lose these free marks

- “Give your answer to an appropriate number of significant figures” means: match the **least** number of significant figures given anywhere in the question's data (never more precise than your least precise input).
- Enthalpy answers always need a **sign** (+ or –) as a distinct, separately awarded mark — even when the context makes exothermic/endothermic “obvious”.
- Units must be stated and must be **consistent** with the numbers used (e.g. do not calculate using J but then quote the final answer as if it were in kJ without converting).

CHAPTER 14

Mastering the 6-Mark (QER) Question

The single most predictable question on the paper — decoded from all 18 sessions

Every paper studied contains exactly one **quality of extended response (QER)** question, marked with an asterisk (*) next to the question number, worth 6 marks. It is marked completely differently from every other question on the paper, and once you understand the mechanics of how it is marked, it becomes one of the most reliably scorable parts of the whole exam — because the content required is almost always a short, learnable list of chemistry points, not a single 'correct answer' to guess.

14.1 How a QER answer is actually marked

Examiners mark a QER answer in two completely separate passes, then add the two marks together:

Component	What is being counted	Typical maximum
1. Indicative content	The examiner has a checklist of ~5–6 specific chemistry points for that exact question. Your raw mark here depends only on how many of these points you have made anywhere in your answer — order and phrasing do not matter, only chemical correctness.	3–4 marks
2. Structure & lines of reasoning	A completely separate judgement of how well-connected and logically sequenced your answer is — do your points link to each other with words like 'because', 'therefore', 'this means that', rather than being a disconnected list of isolated facts?	2 marks

The two are combined using a table that appears, in near-identical form, on every mark scheme studied — broadly:

Indicative points made	Content mark	Structure of answer	Reasoning mark
5–6 points	3–4	Coherent, logical, fully-linked throughout	2
3–4 points	2	Partially structured, some linkage	1
1–2 points	1	No linkage, just a list of disconnected facts	0
0 points	0	—	0

♦ **EXAM TIP — The two practical conclusions that matter most**

- **Coverage beats elegance.** Getting 5–6 separate, chemically correct points onto the page — even as a slightly rough list — already secures most of the marks. A beautifully written paragraph containing only 2 correct chemistry points will always score lower.
- **Then spend your last 30 seconds only on joining words.** Once you have your list of points, re-read it and insert connectives ('because', 'this means', 'as a result', 'therefore', 'in contrast') between them — this single edit is often worth the final 1–2 marks and costs almost no extra chemistry.

⚠ **COMMON ERROR**

If your answer contains a **chemically incorrect** statement, marks are deducted from the *reasoning* mark specifically (not necessarily the content mark) — so a wrong statement tacked onto an otherwise good answer can cost you the structure marks even though your correct points still count. It is safer to leave out a point you are unsure of than to state it wrongly.

14.2 A 5-step method for any QER question

1. **Underline the command** (discuss / explain / compare and contrast / evaluate / describe how) and the exact scope (which species, which two conditions, which two groups) — QER questions often explicitly list sub-requirements as bullet points; treat each one as a mini indicative-content target.
2. **Brain-dump a bullet list first**, in rough order, before writing full sentences — aim for 5–6 separate, self-contained chemistry facts.
3. **Check you have covered ‘both sides’** if the question says compare/contrast, or discuss more than one species/process — a very common way to lose marks is writing five good points about only one side of a comparison.
4. **Write it up as connected prose**, turning your bullet list into 4–6 sentences joined with reasoning words.
5. **Include at least one equation** if the question mentions or implies specific reactions — a correctly balanced equation is almost always one of the indicative content points on offer.

14.3 The recurring QER archetypes — learn these content templates

Across the 18 papers studied, the 6-mark question is drawn from a small, repeating set of themes. Learn the content template for each — the exact substances named will change from session to session, but the chemistry points required do not.

★ 6-MARK QUESTION — Archetype A — “Explain the boiling-temperature pattern of these molecules”

Seen with: isoelectronic hydrides (CH_4 , NH_3 , H_2O , HF); small molecules of similar Mr but different polarity; hydrogen vs ammonia liquefying; halogens/hydrogen halides down a group.

Content template:

1. State which intermolecular force(s) each substance has (all have London forces; identify which also have permanent dipole-dipole and/or hydrogen bonding).
2. For substances with only London forces, more electrons → stronger London forces → use this to explain any difference *within* that subgroup.
3. State that hydrogen bonding is much the strongest of the three forces.
4. Identify which substance(s) can hydrogen bond (H directly bonded to N, O or F) and conclude they have anomalously high boiling temperature(s) despite sometimes having fewer electrons than a rival.
5. If two hydrogen-bonding substances are compared to each other, the tie-breaker is normally the **number of hydrogen bonds each molecule can form** (e.g. H_2O can form more hydrogen bonds per molecule than HF).

★ 6-MARK QUESTION — Archetype B — “Explain why these conditions are used in this industrial process”

Seen with: ethene + steam → ethanol (H_3PO_4 catalyst, 300 °C, 60 atm); the Haber process; methanol synthesis from $\text{CO}_2 + \text{H}_2$; the Sabatier process; the Ostwald process.

Content template (for an exothermic forward reaction with fewer gas moles on the product side):

1. A **lower temperature** would increase the equilibrium yield, because the forward reaction is exothermic (Le Chatelier — shifts right to oppose the removal of heat).
2. But a lower temperature gives a **slower rate** — so the actual temperature used is a **compromise** between rate and yield.
3. **Higher pressure** increases both rate (more frequent collisions) *and* yield (shifts equilibrium towards the side with fewer gas molecules).
4. But very high pressure is expensive — stronger/thicker equipment and higher running (energy) costs — so again a **compromise** pressure is chosen.
5. A **catalyst** is used to increase the rate without needing to raise the temperature (protecting yield), and unreacted gases are **recycled**, so even a modest % yield per single pass is still economically acceptable.

★ 6-MARK QUESTION — Archetype C — “Compare and contrast substitution vs elimination of a halogenoalkane”

Seen with: 2-bromobutane, 2-chlorobutane, etc. + KOH.

Content template:

- Both reactions involve the hydroxide ion, OH^- , and require heating.
- In **aqueous** solution the reaction is **substitution**; OH^- acts as a **nucleophile**; the product is an alcohol — give the equation.
- In **ethanolic** solution the reaction is **elimination**; OH^- acts as a **base**; the product is an alkene — give the equation.
- If more than one alkene is structurally possible, name/draw them, and note that one of them may show E/Z (geometric) isomerism while a structural isomer of it may not.

★ 6-MARK QUESTION — Archetype D — “Compare/evaluate the reactions of concentrated H_2SO_4 with two different halides”

Seen with: chloride vs bromide vs iodide; deciding which halide is the stronger reducing agent from observations.

Content template:

- Both/all halides first undergo simple acid-base (proton-transfer, non-redox) reactions, releasing misty/steamy HX fumes — give this equation.
- State which further redox product(s) are observed for each halide (Br_2 /orange vapour and SO_2 for bromide; I_2 /purple vapour and S and/or H_2S for iodide) — chloride shows no further reaction.
- Give the oxidation number change of sulfur in each case (e.g. $+6 \rightarrow +4$ for SO_2 , $+6 \rightarrow 0$ or -2 for S/ H_2S) to prove which halide is the stronger reducing agent (the one that reduces sulfur furthest).
- Link this to structure: the larger halide ion loses its outer electron more easily (more shielding/further from the nucleus), so is the better reducing agent — and give an ionic equation for the reaction producing the strongest evidence (e.g. the ‘rotten eggs’ smell).

★ 6-MARK QUESTION — Archetype E — “Explain why Group 1 element X behaves like Group 2” / Group 1 vs Group 2 comparisons

Seen with: lithium nitrate decomposing like Group 2; comparing calcium and magnesium's reactions with water.

Content template (lithium/Group 2 similarity):

- Li^+ and Mg^{2+} have similar ionic radii (Li^+ is unusually small for Group 1).
- Give the equation for thermal decomposition of the Group 1/lithium nitrate and of the Group 2 nitrate, showing both decompose to the oxide + NO_2 + O_2 , unlike sodium nitrate (which only forms the nitrite + O_2) — give that equation too, for contrast.
- Explain that the small, high-charge-density $\text{Li}^+/\text{Mg}^{2+}$ ion polarises the nitrate ion more strongly, weakening the N-O bond enough for full decomposition.

★ 6-MARK QUESTION — Archetype F — “Devise an experiment to find the identity/ A_r of an unknown element”

Seen with: finding the relative atomic mass of a Group 1 or Group 2 metal or its carbonate using only simple apparatus (test tube, Bunsen, balance).

Content template:

- Weigh the empty test tube/container, then weigh again after adding a sample of the solid.
- Heat (to constant mass) until the mass no longer changes, to ensure the reaction/decomposition is complete.
- Calculate the moles of gas lost (e.g. CO_2) from the mass loss, using its molar mass.
- Use these moles (equal to the moles of the original compound, from the equation's ratio) together with the initial mass to calculate the molar mass of the compound, then subtract the known mass of the non-metal part to find the A_r of the unknown element, and hence identify it from the Data Booklet.

14.4 A final structural tip

If a QER question gives you bullet-pointed sub-requirements in the stem (“in your answer you should ...”), those bullets are **almost always a direct map of the indicative content points** — answer them in the order given, one clear point per bullet, and you will naturally hit close to the maximum content mark.

CHAPTER 15

Final Pre-Exam Checklist

One self-test pass through the entire specification before you walk in

Go through every line below and be honest with yourself — if you hesitate on any one, open the relevant chapter and re-read it before moving on. Every single item here has been drawn directly from a recurring pattern across the 18 papers analysed for this guide.

Energetics (Ch. 3)

- ☐ I can state all five/six enthalpy definitions ($\Delta_f H$, $\Delta_c H$, $\Delta_a H$, $\Delta_{neut} H$, $\Delta_r H$, $\Delta_{soln} H$) exactly, including 'one mole' and standard conditions.
- ☐ I can construct and use a Hess cycle in either direction (formation data or combustion data), including getting the sign right.
- ☐ I can state both reasons bond-enthalpy answers are less reliable (averaged values; gas phase only).
- ☐ I can do a full $Q = mc\Delta T$ calorimetry calculation, including significant figures, sign and units.
- ☐ I can give at least three non-'heat loss' sources of experimental error in a combustion experiment.
- ☐ I can draw and correctly label a reaction profile, catalysed and uncatalysed, with E_a and ΔH in the right places.

Kinetics & Equilibrium (Ch. 4-5)

- ☐ I can sketch a Maxwell-Boltzmann curve from scratch with all five required features.
- ☐ I can explain, in the full three-link chain, why temperature increases rate.
- ☐ I can explain how a catalyst works using the Maxwell-Boltzmann diagram, and give sustainability reasons for using one.
- ☐ I can find a rate from a tangent on a graph, with correct units, and know I must show the construction lines.
- ☐ I can apply Le Chatelier's Principle to concentration, pressure and temperature changes, and state the effect on rate separately from the effect on yield.
- ☐ I can write the full 'compromise conditions' argument (rate vs yield vs cost) for an industrial equilibrium process.

Redox & Groups 1/2/7 (Ch. 6-8)

- ☐ I know all the oxidation number rules, including the H_2O_2 and OF_2 exceptions.
- ☐ I can identify a disproportionation reaction and state the before/after oxidation numbers for both changes.
- ☐ I can construct and combine half-equations.
- ☐ I know all eight flame test colours (Li, Na, K, Rb, Cs, Ca, Sr, Ba).
- ☐ I can give the full polarisation explanation for Group 2 thermal stability, without reversing cation/anion.
- ☐ I know the Group 2 hydroxide (more soluble down) and sulfate (less soluble down) solubility trends.
- ☐ I know all Group 1/2/lithium nitrate and carbonate decomposition equations and the Group 1 vs Li/Group 2 exception.
- ☐ I can give the full concentrated- H_2SO_4 -with-halide table (products, colours, redox) for Cl^- , Br^- and I^- .
- ☐ I know the silver nitrate precipitate colours and their solubility in ammonia for Cl^- , Br^- , I^- .
- ☐ I know the disproportionation equations for chlorine with cold/hot NaOH and with water.

Intermolecular Forces, Halogenoalkanes & Alcohols (Ch. 9-11)

- ☐ I can state the three hydrogen-bonding criteria and identify which functional groups can/cannot hydrogen bond.

- ☐ I can write the full boiling-temperature explanation template (forces present → electrons/London → strongest force → conclusion).
- ☐ I can draw both the SN1 (tertiary) and SN2 (primary) mechanisms fully, with correct arrows, dipoles and lone pairs.
- ☐ I can explain the hydrolysis rate trend ($\text{Cl} < \text{Br} < \text{I}$) using bond enthalpy — never electronegativity.
- ☐ I know which conditions give substitution vs elimination with KOH, and can draw the elimination mechanism.
- ☐ I know the reagents/conditions for halogenoalkane → alcohol, nitrile, amine and alkene.
- ☐ I know which alcohol type oxidises to what, the exact apparatus (distil vs reflux) for each, and the orange→green colour change.
- ☐ I know the reagents to convert an alcohol into a chloro-, bromo- and iodo-alkane, and the PCl_5 test for $-\text{OH}$.

Spectroscopy & Practical Skills (Ch. 12-13)

- ☐ I know the wavenumber range and bond for O-H (alcohol), O-H (acid), N-H, $\text{C}\equiv\text{N}$, $\text{C}=\text{O}$ and $\text{C}=\text{C}$.
- ☐ I can identify the molecular ion peak and assign at least the common fragment ions (15, 29, 43, 45).
- ☐ I can carry out a full titration calculation, including scaling from a volumetric flask sample.
- ☐ I know the concordant-titre rule and how to calculate percentage uncertainty in a titre.
- ☐ I can do an empirical/molecular formula calculation from mass or percentage composition data.
- ☐ I can use $pV = nRT$ with all three unit conversions ($\text{kPa}\rightarrow\text{Pa}$, $\text{cm}^3\rightarrow\text{m}^3$, $^\circ\text{C}\rightarrow\text{K}$) done correctly.

Exam Technique (Ch. 1, 14)

- ☐ I never leave a calculation blank, even if I am unsure — partial working always scores partial credit.
- ☐ I always include state symbols when asked, and always give sign, units and appropriate significant figures.
- ☐ For any 6-mark (*) question, I write a bullet list of 5–6 separate chemistry points first, then link them with reasoning words afterwards.
- ☐ I recognise the recurring QER archetypes (boiling-temperature explanation; industrial conditions; substitution vs elimination; halide/ H_2SO_4 comparison; Group 1/2 comparison; devise an experiment) and know the content template for each.

♦ EXAM TIP — Last word

Every pattern in this guide was drawn from real, repeated marking points across 18 consecutive sittings of this exact paper. Nothing here is guesswork — if a topic appears in this guide, it is because it appeared, in some form, in the real assessment record. Work through it once properly, use this checklist to find your gaps, and go back into the relevant chapter for anything you are not yet completely confident on. Good luck.